Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Gary Pellandini

Physical address of dairy:

Number and Street	City	County	Zip Code
13020 Pellandini RD	Galt	Sacramento	95632

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

0146-0230-0020-0000 0146-0230-0021-0000 0146-0230-0022-0000

B. OPERATORS

Pellandini, Gary

Operator name: Pellandini, Gary

Telephone no.: (209) 745-1488 Landline (209) 810-8545 Cellular

Mailing Address Number and Street	City	State	Zip Code
13020 Pellandini RD	Galt	CA	95632

This operator is responsible for paying permit fees.

C. OWNERS

Pellandini, Gary

Legal owner name: Pellandini, Gary

Telephone no.: (209) 745-1488 Landline (209) 810-8545 Cellular

Mailing Address Number and Street	City	State	Zip Code
13020 Pellandini RD	Galt	CA	95632

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #3 - Kull Dairy	Ground water
Ag Well #5 - 35-80	Ground water
Ag Well #6 - Keller	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/01/2023	Liquid commercial fertilizer *UN32*	13,600.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	36,317.44	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	36,317.44	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
HP-1	40	40	0	none	X146-0230-X021-0000
HP-2	112	112	1	none	X146-0230-X021-0000
HP-3	50	50	0	none	X146-0230-X022-0000
HP-4	147	147	1	none	X146-0230-X022-0000
HP-5	47	47	1	none	X146-0230-X020-0000
Totals for areas that were used for application					
Totals for areas that were not used for application	396	396	3		
Land application area totals	396	396	3		

B. CROPS AND HARVESTS

HP-2

Field name: HP-2

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 82 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	2,120.00 *ton*	Dry-weight		68.0	12,200.00	1,470.00	6,350.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	25.85	201.87	24.32	105.07	827.32

HP-4

Field name: HP-4

HP-4

11/01/2018: Almond, in shell

Crop: Almond, in shell Acres planted: 79 Plant date: 11/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	79.00 *ton*	Dry-weight		10.0	37,100.00	1,860.00	37,300.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.00	130.00	20.00	100.00	0.00
Total actual harvest content	1.00	66.78	3.35	67.14	163.80

HP-5

Field name: HP-5

01/01/2015: Almond, in shell

Crop: Almond, in shell Acres planted: 47 Plant date: 01/01/2015

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	52.00 *ton*	Dry-weight		11.0	37,100.00	1,860.00	37,300.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.00	130.00	20.00	100.00	0.00
Total actual harvest content	1.11	73.06	3.66	73.46	179.21

NUTRIENT BUDGET

A. LAND APPLICATIONS

HP-2 - 06/28/2023: Corn, silage

Field name: HP-2

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	8,995,550.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	4,169,920.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	9,056,375.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	245.00	0.00	0.00	0.00	
Application event totals		245.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	10,689,975.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

HP-2 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	210.29	10,596,500.00 *gal*
Application event totals		0.00	0.00	0.00	210.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	10,861,200.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	11,983,500.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #5 - 35-80	Ground water	0.00	0.00	0.00	0.00	10,617,500.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

HP-4 - 11/01/2018: Almond, in shell

Field name: HP-4

Crop: Almond, in shell

Plant date: 11/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3 - Kull Dairy	Ground water	0.00	0.00	0.00	0.00	51,484,525.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

HP-4 - 11/01/2018: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	140.00	0.00	0.00	0.00	
Application event totals		140.00	0.00	0.00	0.00	

HP-5 - 01/01/2015: Almond, in shell

Field name: HP-5

Crop: Almond, in shell

Plant date: 01/01/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.03	0.00	0.00	0.00	390,000.00 *gal*
Application event totals		0.03	0.00	0.00	0.00	

05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.08	0.00	0.00	0.00	1,185,000.00 *gal*
Application event totals		0.08	0.00	0.00	0.00	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.06	0.00	0.00	0.00	915,000.00 *gal*
Application event totals		0.06	0.00	0.00	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.07	0.00	0.00	0.00	1,020,000.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

HP-5 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.04	0.00	0.00	0.00	540,000.00 *gal*
Application event totals		0.04	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	35.00	0.00	0.00	0.00	
Application event totals		35.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.06	0.00	0.00	0.00	864,500.00 *gal*
Application event totals		0.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.09	0.00	0.00	0.00	1,339,500.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.07	0.00	0.00	0.00	1,111,500.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.09	0.00	0.00	0.00	1,282,500.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

HP-5 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.03	0.00	0.00	0.00	471,750.00 *gal*
Application event totals		0.03	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.09	0.00	0.00	0.00	1,339,200.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	40.00	0.00	0.00	0.00	
Application event totals		40.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.05	0.00	0.00	0.00	670,625.00 *gal*
Application event totals		0.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.02	0.00	0.00	0.00	300,750.00 *gal*
Application event totals		0.02	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.04	0.00	0.00	0.00	608,000.00 *gal*
Application event totals		0.04	0.00	0.00	0.00	

HP-5 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.05	0.00	0.00	0.00	783,750.00 *gal*
Application event totals		0.05	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.05	0.00	0.00	0.00	747,000.00 *gal*
Application event totals		0.05	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.06	0.00	0.00	0.00	832,650.00 *gal*
Application event totals		0.06	0.00	0.00	0.00	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.07	0.00	0.00	0.00	1,070,160.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.08	0.00	0.00	0.00	1,197,000.00 *gal*
Application event totals		0.08	0.00	0.00	0.00	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.06	0.00	0.00	0.00	874,000.00 *gal*
Application event totals		0.06	0.00	0.00	0.00	

HP-5 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.06	0.00	0.00	0.00	860,250.00 *gal*
Application event totals		0.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.09	0.00	0.00	0.00	1,310,400.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.05	0.00	0.00	0.00	679,875.00 *gal*
Application event totals		0.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.03	0.00	0.00	0.00	382,200.00 *gal*
Application event totals		0.03	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.02	0.00	0.00	0.00	368,550.00 *gal*
Application event totals		0.02	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.03	0.00	0.00	0.00	441,000.00 *gal*
Application event totals		0.03	0.00	0.00	0.00	

HP-5 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.06	0.00	0.00	0.00	960,300.00 *gal*
Application event totals		0.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	35.00	0.00	0.00	0.00	
Application event totals		35.00	0.00	0.00	0.00	

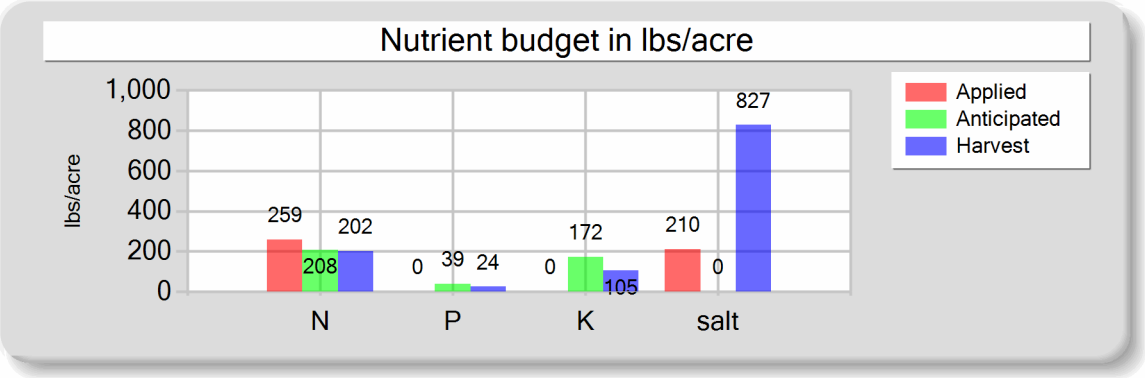
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #6 - Keller	Ground water	0.04	0.00	0.00	0.00	585,000.00 *gal*
Application event totals		0.04	0.00	0.00	0.00	

B. NUTRIENT BUDGET

HP-2 - 06/28/2023: Corn, silage

Field name: HP-2 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	245.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	210.29
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	259.00	0.00	0.00	210.29
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	201.87	24.32	105.07	827.32
Nutrient balance	57.13	-24.32	-105.07	-617.03
Applied to removed ratio	1.28	0.00	0.00	0.25

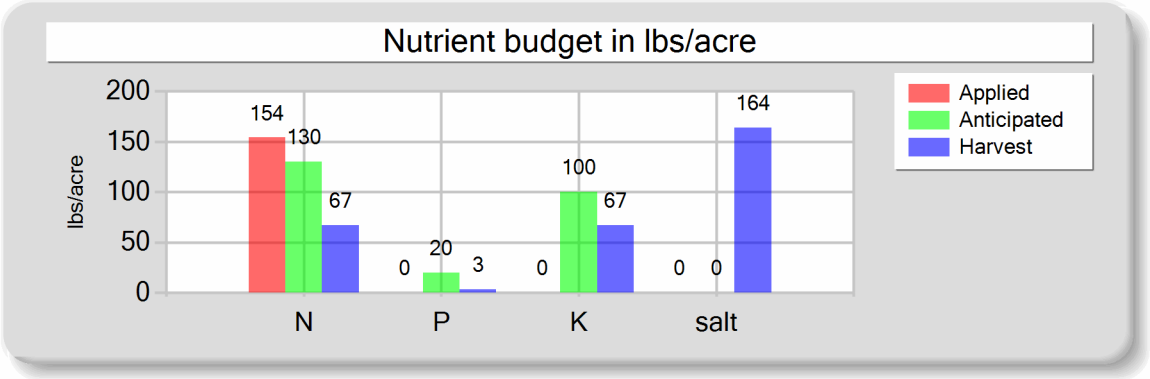
Fresh water applied	
76,970,520.00	*gallons*
2,834.56	*acre-inches*
34.57	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

HP-4 - 11/01/2018: Almond, in shell

Field name: HP-4 Crop: Almond, in shell Plant date: 11/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	140.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	154.00	0.00	0.00	0.00
Anticipated crop nutrient removal	130.00	20.00	100.00	0.00
Actual crop nutrient removal	66.78	3.35	67.14	163.80
Nutrient balance	87.22	-3.35	-67.14	-163.80
Applied to removed ratio	2.31	0.00	0.00	0.00

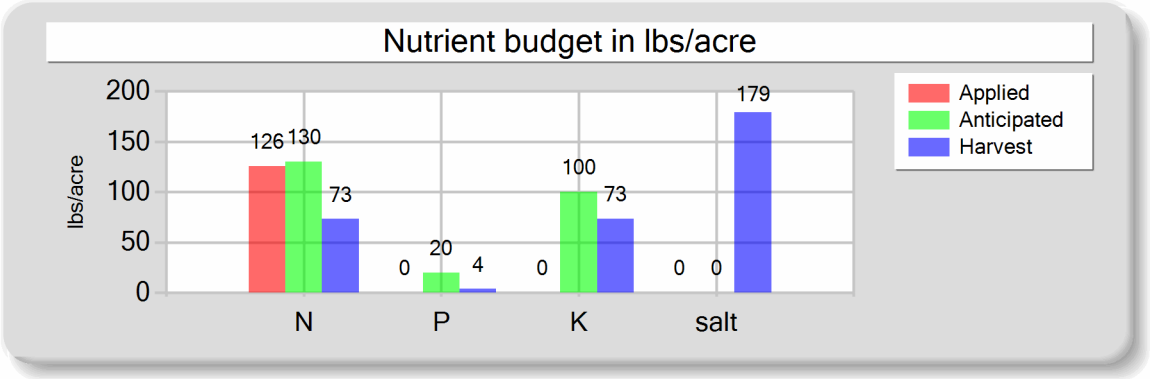
Fresh water applied	
51,484,525.00	*gallons*
1,896.00	*acre-inches*
24.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

HP-5 - 01/01/2015: Almond, in shell

Field name: HP-5 Crop: Almond, in shell Plant date: 01/01/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	110.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.56	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	125.56	0.00	0.00	0.00
Anticipated crop nutrient removal	130.00	20.00	100.00	0.00
Actual crop nutrient removal	73.06	3.66	73.46	179.21
Nutrient balance	52.50	-3.66	-73.46	-179.21
Applied to removed ratio	1.72	0.00	0.00	0.00

Fresh water applied	
23,130,460.00	*gallons*
851.82	*acre-inches*
18.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

Ag Well #3 - Kull Dairy

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 10/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	11.60	5.90	30.10	262.00	0.00	18.10	5.00	385.00	
DL	0.14	0.50	0.14	0.10	0.10	0.10	1.00	1.00	0.10	0.10	1.00	

Ag Well #5 - 35-80

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 10/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	11.60	6.10	22.30	220.00	0.00	10.50	8.40	325.00	
DL	0.14	0.50	0.14	0.10	0.10	0.10	1.00	1.00	0.10	0.10	1.00	

Ag Well #6 - Keller

Ag Well #6 - Keller

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 10/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.38	0.00	0.38	45.50	24.30	26.50	397.00	0.00	23.20	26.00	692.00	
DL	0.10	0.50	0.10	0.10	0.10	0.10	0.10	1.00	0.10	0.10	1.00	

D. SOIL ANALYSES

HP-1

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 04/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			7.00				
DL			0.10				

HP-2

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 04/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			8.00				
DL			0.10				

HP-3

HP-3

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 04/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			6.00				
DL			0.10				

HP-4

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 04/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			1.00				

HP-5

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 04/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3.00				
DL			0.10				

E. PLANT TISSUE ANALYSES

HP-2 - 06/28/2023: Corn, silage

HP-2 - 06/28/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	1,470.00	6,350.00		5.00
DL	100.00	10.00	10.00		0.10

HP-4 - 11/01/2018: Almond, in shell

Estimate - Not Sampled

Sample and source description: Estimate - Not Sampled

Sample date: 11/01/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 11.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,100.00	1,470.00	37,300.00		9.10
DL	100.00	10.00	100.00		0.10

HP-5 - 01/01/2015: Almond, in shell

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 04/08/2024 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,100.00	1,860.00	37,300.00		9.10
DL	100.00	10.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

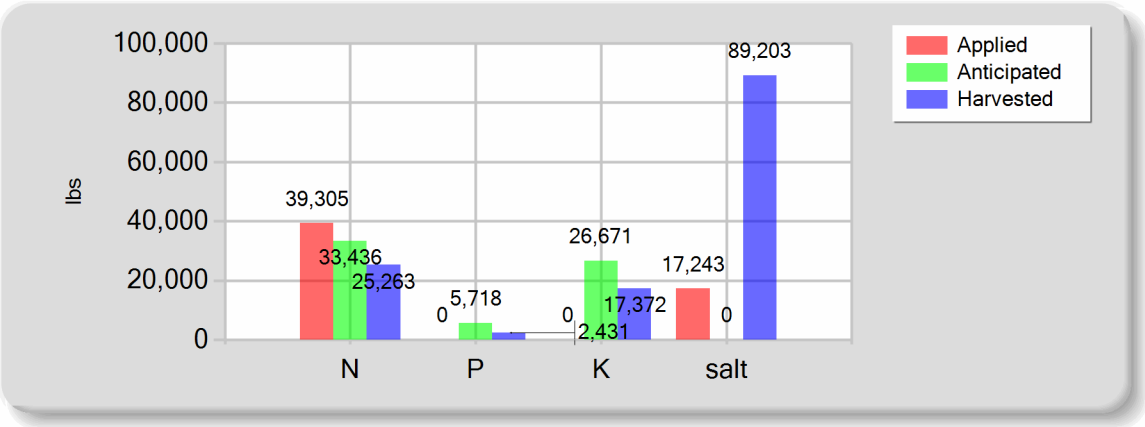
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

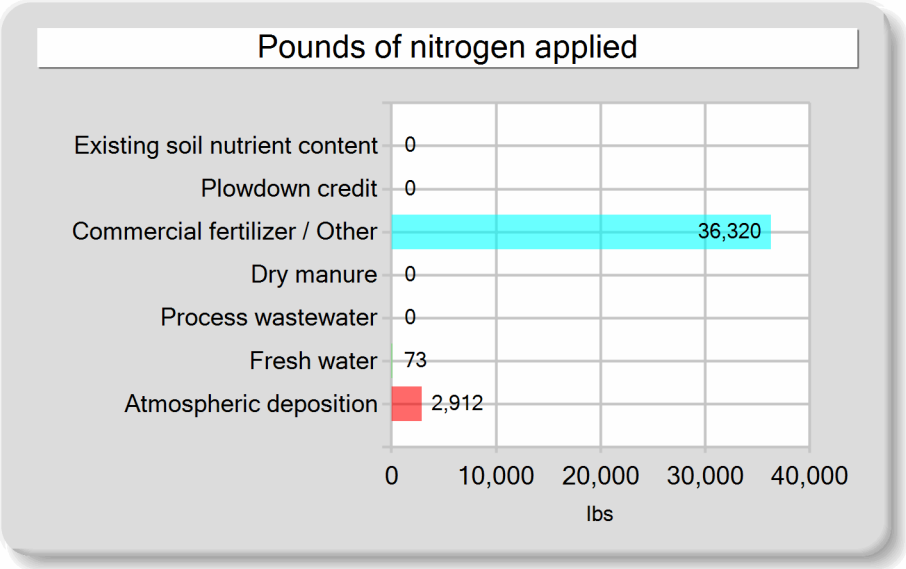
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

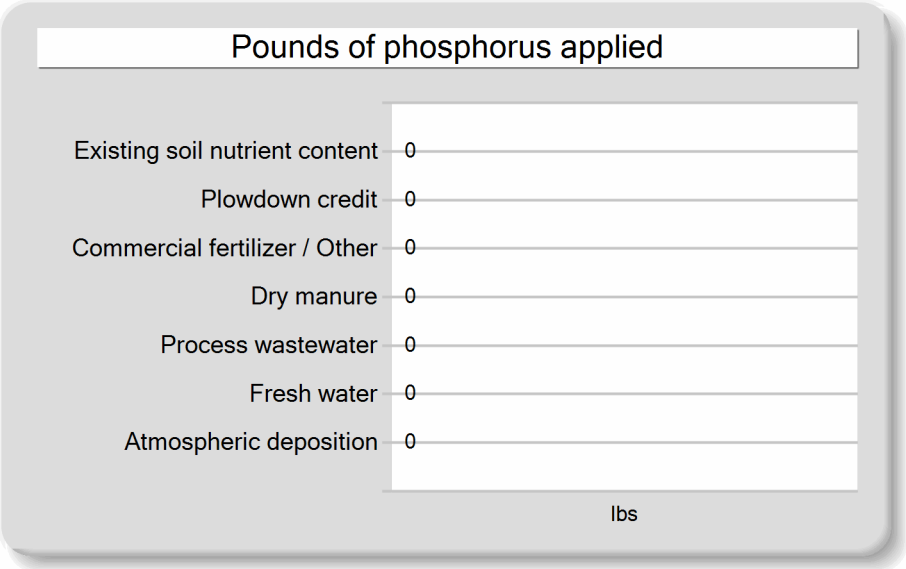
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	36,320.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	73.35	0.00	0.00	17,243.42
Atmospheric deposition	2,912.00	0.00	0.00	0.00
Total nutrients applied	39,305.35	0.00	0.00	17,243.42
Anticipated crop nutrient removal	33,436.00	5,718.00	26,671.20	0.00
Actual crop nutrient removal	25,262.56	2,431.15	17,372.23	89,203.16
Nutrient balance	14,042.79	-2,431.15	-17,372.23	-71,959.74
Applied to removed ratio	1.56	0.00	0.00	0.19

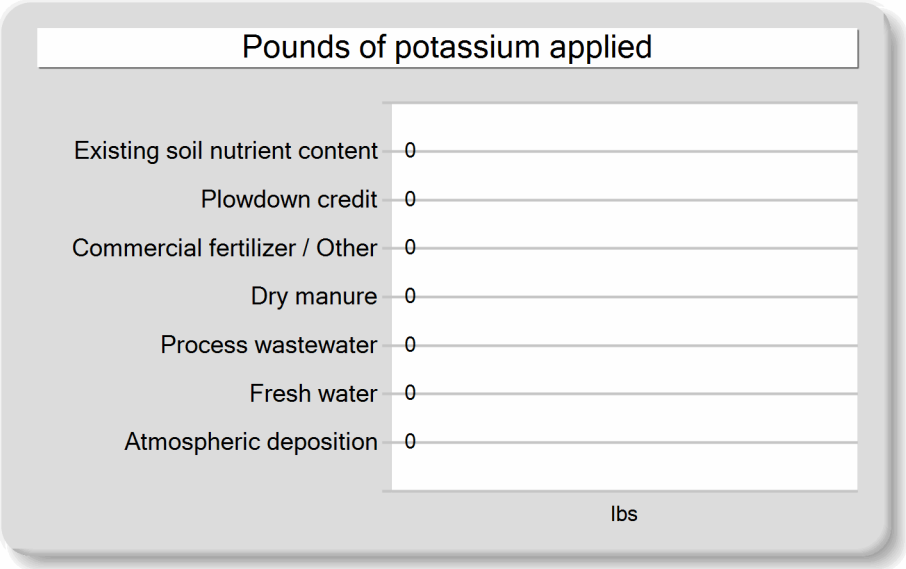
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

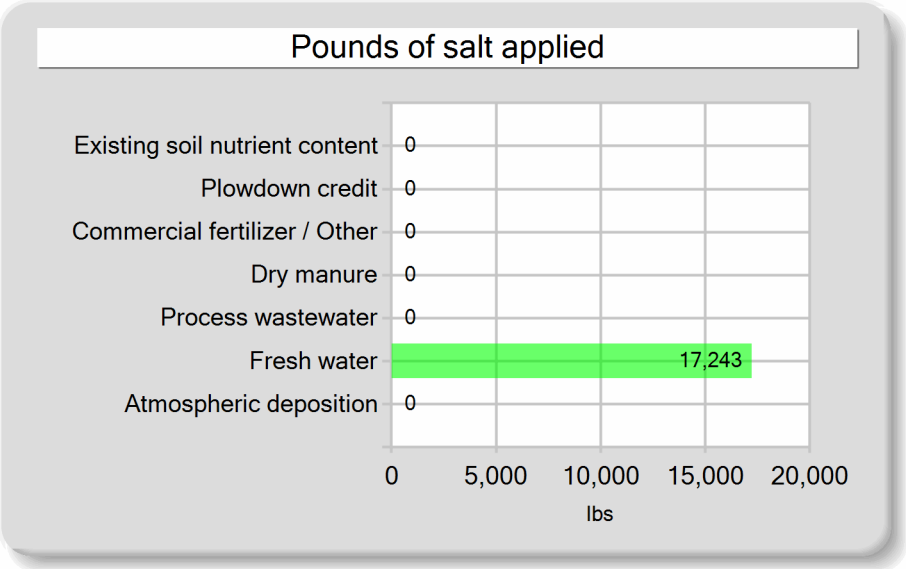


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

There are no cattle on the dairy and no manure remains. Also no process waste water.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Gary Pellandini	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Gary Pellandini	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Gary Pellandini	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/26/24	
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 10/23/23 11:10 AM
SUBMITTED BY: Gary
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Keller	T3296042A	Electrical Conductivity (EC)	EPA 120.1 and	0.692	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.386	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	443	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	97.0	98.7 (T3296043A)	100 (T3296043A)	1.53

Water Orchard Pellandini ~ Robson Rd

Fields 17
18
19
20

T arranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the c ner uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

Agricultural and Environmental Analysis **ELAP Certificate No. 2714**

Water Analysis

Date 10/23/2023 11:18 AM
Report Date: 10/24/2023 4:03 PM
Lab ID: **T3296043A**

Source: Irrigation Water
Purchase Order:

Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Sample ID: Keller Ag Well
Submitted By: Gary
Grower:
PCA:

Analyte	Result	Units	meq/L	lbs/ac in	lbs/ac ft	Low	Moderate	High
pH	7.7	Units						
Electrical Conductivity	0.692	mmhos/cm						
Soluble Salts	443	mg/L						
Nitrate Nitrogen	< 0.140	ppm						
Nitrate	0.620	ppm		0.140	1.67			
Bicarbonate	397	ppm	6.50	89.2	1070			
Calcium	45.5	ppm	2.28	10.2	123			
Magnesium	24.3	ppm	2.00	5.46	65.6			
Sodium	26.5	ppm	1.15	5.95	71.4			
Potassium	2.37	ppm	0.061	0.533	6.40			
Boron	0.070	ppm		0.016	0.189			
Chloride	26.0	ppm	0.732	5.85	70.2			
Sulfate	23.2	ppm		5.22	62.7			
Sodium Absorption Ratio	0.787							

Lbs. of N from this water at 36 in per ac/year: < 1.13
Lbs. of B from this water at 36 in per ac/year: 0.567

Reviewed and approved by: _____

Orchard Pellandini and Robson Rd
Fields 17
18
19
20

If QC is required for this sample, please contact lab.

Liability Limits: The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, expressed or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Turlock, CA Ph: (209) 634-9055
Woodland, CA Ph: (530) 666-9056
www.denelelabs.com Fax: (209) 634-9057

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Compliance Analysis Report

Customer
Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 10/23/23 11:10 AM
SUBMITTED BY: Gary
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
35-80	T3296042B	Electrical Conductivity (EC)	EPA 120.1 and	0.325	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	208	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	97.0	98.7 (T3296043A)	100 (T3296043A)	1.53

Water Source for Fields

11

12

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

Agricultural and Environmental Analysis

ELAP Certificate No. 2714

Water Analysis

Date 10/23/2023 11:18 AM

Report Date: 10/24/2023 4:03 PM

Lab ID: **T3296043B**

Source: Irrigation Water

Purchase Order:

Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Sample ID: 3580 Ag Well
Submitted By: Gary
Grower:
PCA:

Analyte	Result	Units	meq/L	lbs/ac in	lbs/ac ft	Low	Moderate	High
pH	8.0	Units						
Electrical Conductivity	0.325	mmhos/cm						
Soluble Salts	208	mg/L						
Nitrate Nitrogen	< 0.140	ppm						
Nitrate	< 0.600	ppm						
Bicarbonate	220	ppm	3.60	49.4	593			
Calcium	11.6	ppm	0.579	2.60	31.2			
Magnesium	6.13	ppm	0.505	1.38	16.6			
Sodium	22.3	ppm	0.970	5.02	60.2			
Potassium	1.03	ppm	0.026	0.232	2.78			
Boron	0.110	ppm		0.025	0.297			
Chloride	8.45	ppm	0.238	1.90	22.8			
fate	10.5	ppm		2.37	28.4			
Sodium Absorption Ratio	1.32							

Lbs. of N from this water at 36 in per ac/year: < 1.13

Lbs. of B from this water at 36 in per ac/year: 0.891

Reviewed and approved by: _____

Water Source for Fields
11
12

If QC is required for this sample, please contact lab.

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Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 10/23/23 11:10 AM
SUBMITTED BY: Gary
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Kuil Dairy	T3296042C	Electrical Conductivity (EC)	EPA 120.1 and	0.385	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	246	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	97.0	98.7 (T3296043A)	100 (T3296043A)	1.53

Water Source For Fields

1 6
2 7
3 8
5

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

Agricultural and Environmental Analysis **ELAP Certificate No. 2714**

Water Analysis

Date 10/23/2023 11:18 AM
Report Date: 10/24/2023 4:03 PM
Lab ID: **T3296043C**

Source: Irrigation Water
Purchase Order:

Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Sample ID: **Kuil Dairy Ag Well**
Submitted By: **Gary**
Grower:
PCA:

Analyte	Result	Units	meq/L	lbs/ac in	lbs/ac ft	Low	Moderate	High
pH	8.0	Units						
Electrical Conductivity	0.385	mmhos/cm						
Soluble Salts	246	mg/L						
Nitrate Nitrogen	< 0.140	ppm						
Nitrate	< 0.600	ppm						
Bicarbonate	262	ppm	4.30	59.0	708			
Calcium	11.6	ppm	0.582	2.62	31.4			
Magnesium	5.89	ppm	0.485	1.33	15.9			
Sodium	30.1	ppm	1.31	6.77	81.2			
Potassium	0.970	ppm	0.025	0.218	2.62			
Boron	0.200	ppm		0.045	0.540			
Chloride	5.00	ppm	0.141	1.13	13.5			
lfate	18.1	ppm		4.06	48.7			
Sodium Absorption Ratio	1.79							

Lbs. of N from this water at 36 in per ac/year: < 1.13
Lbs. of B from this water at 36 in per ac/year: 1.62

Reviewed and approved by: _____

Water Source For Fields

1 5
2 6
3 7
8

If QC is required for this sample, please contact lab

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Compliance Analysis Report

Customer
Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 10/23/23 11:10 AM
SUBMITTED BY: Gary
ANALYZED DATE: 10/23/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well Dairy	T3296042D	Electrical Conductivity (EC)	EPA 120.1 and	1.04	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.01	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	668	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.4	96.4 (T3293019G)	97.6 (T3293019G)	1.23
Electrical Conductivity (EC)	94.3			1.20
Nitrate Nitrogen (NO3-N)	97.0	98.7 (T3296043A)	100 (T3296043A)	1.53

Domestic Water Well

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

Agricultural and Environmental Analysis

ELAP Certificate No. 2714

Water Analysis

Date 10/23/2023 11:18 AM
Report Date: 10/24/2023 4:03 PM
Lab ID: **T3296043D**

Source: Irrigation Water
Purchase Order:

Gary Pellandini Ranch
13020 Pellandini Rd.
Galt, CA 95632-8818

Sample ID: Domestic Well Dairy
Submitted By: Gary
Grower:
PCA:

Analyte	Result	Units	meq/L	lbs/ac in	lbs/ac ft	Low	Moderate	High
pH	7.7	Units						
Electrical Conductivity	1.24	mmhos/cm						
Soluble Salts	796	mg/L						
Nitrate Nitrogen	3.01	ppm		0.677	8.12			
Nitrate	13.3	ppm		3.00	36.0			
Bicarbonate	641	ppm	10.5	144	1730			
Calcium	97.1	ppm	4.85	21.8	262			
Magnesium	57.1	ppm	4.70	12.8	154			
Sodium	46.5	ppm	2.02	10.5	126			
Potassium	4.66	ppm	0.119	1.05	12.6			
Boron	0.070	ppm		0.016	0.189			
Chloride	43.1	ppm	1.21	9.69	116			
Sulfate	34.5	ppm		7.76	93.2			
Sodium Absorption Ratio	0.926							

Lbs. of N from this water at 36 in per ac/year: 24.4
Lbs. of B from this water at 36 in per ac/year: 0.567

Reviewed and approved by: _____

DOMESTIC Water Well

If QC is required for this sample, please contact lab.

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: Dairy Compliance Bruce Adamek

Client: Gary Pellandini
Mailing address: 13020 Pellandini
(not facility address) Galt Ca. 95632
Phone#: 209 810-8545

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10/12/23	5:00 PM	Keller	AG WELL	15.6°		✓				
2	10/6/23	7:45 PM	35-80	AG WELL	15.8°		✓				
3	10/9/23	4:30 AM	Kuil Dairy	AG WELL	15.0°		✓				
4	10/23/23	8:45 AM	Domestic Well dairy	Domestic Well	17.6°		✓				
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	Gary Pellandini		10/23/23 11:10 AM			
2nd		DENELE ANALYTICAL, INC.	10-23-23 11:10			T32910042

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)